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Comparison of Light Scattering Devices and Impactors for Particulate Measurements in Indoor, Outdoor, and Personal Environments

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Short-term monitoring of individual particulate matter (PM) exposures on subjects and inside residences in health effect studies have been sparse due to the lack of adequate monitoring devices. The recent development of small and portable light scattering devices, including the Radiance nephelometer (neph) and the personal DataRAM (pDR) has made this monitoring possible. This paper evaluates the performance of both the passive pDR and neph (without any size fractionation inlet) against measurements from both Harvard impactors ($HI_{2.5}$) and Harvard personal environmental monitors ($HPEM_{2.5}$) for $PM_{2.5}$ in indoor, outdoor, and personal settings. These measurements were taken at the residences and on the person of nonsmoking elderly subjects across the metropolitan Seattle area and represent a wide range of light scattering measurements directly related to exposures and health effects. At low PM levels, nephs provided finer resolution and more precise measurements (precision = 3−8% and uncertainty = 2.8×10^{-7} m^{-1} or <1 $\mu g/m^3$) than the pDRs. The unbiased precision of pDRs above 10 $\mu g/m^3$ is around 5% (with an unbiased uncertainty of 4.4 $\mu g/m^3$). The 24-h average responses of the pDR and neph, as compared to 24-h integrated gravimetric measurements, are not affected by indoor sources of PM. When regressed against 24-h gravimetric measurements, nephs showed higher coefficients of determination (R^2 = 0.81−0.93) than pDRs (R^2 = 0.77−0.84). The default mass calibration on the pDRs generally overestimated indoor $HI_{2.5}$ measurements by 56%. When carried by subjects, the pDR overestimated the $HPEM_{2.5}$ measurements by approximately 27%. Collocated real-time indoor nephs and pDRs at diverse residential sites had varied coefficients of determination across homes (R^2 = 0.75−0.96), and the difference between pDR and neph responses increased during cooking hours. This difference was larger during baking or frying episodes than during other cooking or cleaning activities. Relative humidity, ranging between 25% and 64% indoors in our study, was not a significant factor affecting the differences in neph or pDR response. In summary, for nonsmoking residences, the mass scattering efficiency (m^2/g) of a stationary indoor neph on a 24-h basis does not vary by residence, including residences with and without cooking activities. This is also true for the pDR. These same stationary indoor pDRs and nephs correlate well with each other, even on a 10-min basis, in the absence of indoor source activities. The fact that these activities comprised a relatively small percentage (cooking + cleaning = 2.3%) of the overall sampling time meant that the overall correlation between these two instruments for all time periods was good. However, when examining the cooking and cleaning periods separately, the correlation was not very good. Thus, during these short-term PM episodes, the 24-h average calibrations versus gravimetric mass should be used with caution. Both devices should be potentially useful in future exposure assessment and health effects studies.

Introduction

Although federal air quality standards for particulate matter (PM) in the United States are based on integrated 24-h and annual mass concentrations, recent studies have provided evidence of adverse health effects from short-term exposures to PM (*1*−*4*). These studies used continuous PM measurements with a tapering element oscillation microbalance (TEOM) monitor at centrally located ambient monitoring sites. The relationship between the hourly outdoor measurements and those experienced by susceptible subjects in/near their homes is still unknown. Until recently, the lack of adequate monitoring devices has hampered health effect studies that required short-term measurements of personal or indoor PM exposures. The recent development of light scattering devices made this possible. These devices include the Radiance nephelometer (Radiance Research, Seattle, WA), hereafter referred to as neph, the personal DataRAM or pDR (Thermo-MIE Inc., Smyrna, GA), and the DustTrak aerosol monitor (TSI Inc., St. Paul, MN) (*5*−*9*). Of these light scattering devices, the pDR (in passive mode, without a size fractionation inlet) has been most widely used in exposure and health effect studies due to its light weight (no pump required), data logging capability, and easy handling (*7, 8, 10*−*13*).

Simultaneous measurements of both gravimetric mass and light scattering coefficients have provided assessment of both the neph and pDR in outdoor and indoor air (*5, 7, 14*−*17*). Less information is available to assess their performance in personal environments (*6, 12*). With the increasing application of these light scattering devices in many exposure and health effect studies, it is important to evaluate and characterize their performance, especially in various microenvironments. This paper compares both the pDR and neph against gravimetric measurements from both the Harvard impactor for $PM_{2.5}$ ($HI_{2.5}$) and the Harvard personal environmental monitor ($HPEM_{2.5}$) in indoor, outdoor, and personal settings. These measurements were taken at residences and on the person of elderly subjects living across the metropolitan Seattle area, as part of a larger study that examines PM exposure and health effects (*13*).

Methods

Study Design. This study was conducted between October 1999 and August 2000. Sixty-three elderly subjects residing in 40 homes (29 private and 11 group homes) participated. All subjects were nonsmokers living in smoke-free homes. PM mass concentration and light scattering were monitored at each residence for 10 consecutive days, both inside and

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TABLE 1. Study Design for Monitor Comparison

comparison instrument	Test	location	(a) neph	(b) pDR
HI (24-h)	1	indoor	40 homes, $N = 314$	16 homes, $N = 100$
	2	outdoor	38 homes, $N = 303$	not performed
HPEM (24-h)	3	personal	not performed	16 subjects, $N = 83$
HI and HPEM (24-h)	4	indoor	2 homes, $N = 18$	2 homes, $N = 18$ (same homes as in Test 4a)
Neph (10-min)	5	indoor	insufficient data points and range	(i) 9 homes, $N = 8752$; (ii) 1 home (1 neph vs 3 pDR), $N = 1711$
	6	lab	5 sessions (S3−S7) (up to 9 nephs) $N = 1261$	3 sessions (S5−S7) (3 pDRs vs 9 nephs) $N = 812$
PDR (10-min)	7	indoor	same as in Test 5b	1 home (3 pDRs, same home as in Test 5bii), $N = 1566$
	8	lab	same as in Test 6b	3 sessions (S5−S7) (3 pDRs), $N = 812$

immediately outside each residence. All monitor inlets were ~1 m above the floor indoors and 1.5 m above the ground outdoors. Instruments were placed away from high-intensity light and away from windows, doors, and vents to avoid drafts.

The study design for monitor comparisons is shown in Table 1. One neph was collocated with one 24-h integrated $HI_{2.5}$ (*18*), resulting in 314 data pairs at 40 home indoor sites (Test 1a) and 303 data pairs at 38 home outdoor sites (Test 2a). Neph was not deployed on subjects (Test 3a). At two of the 40 home indoor sites, neph was collocated with an $HI_{2.5}$, $HPEM_{2.5}$ (*19*, *20*), and pDR (Test 4a). The pDR was collocated with the $HI_{2.5}$ at 16 home indoor sites (Test 1b) but was not deployed outdoors. Sixteen subjects carried both the $HPEM_{2.5}$ and the pDR for up to 10 days (Test 3b). The $HPEM_{2.5}$ was worn between the neck and the chest while the pDR was worn near the waist for convenience. Although the pDR weighs only 0.5 kg, its size (153 mm H × 92 mm W × 63 mm D) makes it uncomfortable when worn in the breathing zone. At this time, it is not clear whether monitoring location (chest vs hip or waist) biases personal exposure measurements. All subjects kept a time−activity diary at 15-min intervals and answered a daily PM episode questionnaire administered by technicians that gathered information on potential PM related activities, including smoking, type of cooking, vacuuming, dusting, burning candles, and window opening.

During each monitoring session, the outdoor neph was brought indoors and collocated with the indoor neph for 15−30 min, with readings set at 1-min resolution (Test 5a). While this provided a useful exercise for synchronizing times between the indoor and outdoor nephs, the data points and concentration range were not sufficient for meaningful comparisons between the nephs in the field (Test 5a). As an alternative to test the agreement of nephs, all nephs were collocated on the same benchtop in our laboratory for up to 24 h after five of the monitoring sessions (Test 6a). At nine of the 16 home indoor sites in Test 1b, the neph was also collocated with a pDR (Test 5bi). Precision of the pDR was determined from one 11-day field collocation test at a single residence (Test 7b). This 11-day test involved three collocated pDRs and one neph (also Test 5bii) and examined the effect of zeroing frequency on the pDR response as well as precision. During this 11-day test, one pDR (pDR1) was zeroed every other day, while the other two instruments (pDR2 and pDR3) were zeroed only at the beginning of the test. Results from this 11-day test were later validated by a 24-h follow-up test ($N = 145$), during which two pDRs were zeroed in one zeroing bag while the third pDR was zeroed individually. The precision test was also performed in the lab with three pDRs collocated with nine nephs after three separate monitoring sessions (Test 8b).

Radiance Nephelometer. The neph (models 902 and 903) integrate the scattered light in such a way as to respond to the mass median aerodynamic diameter (d_{p50}) primarily between 0.1 and 1 μm (*15*), with a sharp decrease in light scattering response for $d_p > 1$ μm. The major difference between 902 and 903 models is that 902 neph is an earlier model and could only be calibrated by the manufacturer. The neph measures light scattering coefficient, B_{sp}, with a wavelength-defining optical filter at 530 nm and a variable-rate flashlamp. The instrument has an internal data logger and a real-time clock to store B_{sp}. The neph was programmed to store 10-min average readings. Small fans (0.08 A, 0.96 W; Minispiral MDC, Ametek Rotron TMD, Suagerties, NY) were used with both indoor and outdoor nephs to ensure sufficient air movement. The inlet of the outdoor neph was made of copper tubing and was fitted with a rain cover. There was no size fractionation inlet attached to the neph. The copper tubing was connected to a constant temperature heater (Radiance Research, Seattle, WA) that raised sample air temperature to 10 °C above ambient air. Heated nephelometers with constant airflow effectively adjust the relative humidity (RH) and bring the nephelometry measurements closer to gravimetric $PM_{2.5}$, as the ratio of B_{sp} to gravimetric $PM_{2.5}$ concentration increases exponentially at high RH values, especially when RH > 60% (*7*, *17*). The indoor neph inlet was not heated, but the relative humidity was monitored hourly near the inlet.

Before each 10-day sampling period, all seven model 903 nephs were purged with clean dry air that had passed through a cartridge of silica gel (AquaPrep600, Gelman, Ann Arbor, MI) and two 0.45-μm hydrophilic polyethersulfone filters (Supor, Gelman) connected in series. These nephs were then calibrated in the lab with medical-grade pure CO_2. The two model 902 nephs used in this study (not suitable for lab calibration) showed a measurable zero drift, which was defined as a significant drift of the baseline over time. Whenever collocated measurements for models 902 and 903 nephs were available, model 902 measurements were adjusted to the same mean as the model 903 measurements. This adjustment varied with session, ranging between −0.635 and 3.329 ($\times 10^{-5}$ m^{-1}). In the absence of this adjustment, model 902 measurements were removed from further analysis (~15% of the analyzed neph measurements were from the model 902).

Thermo-MIE Personal DataRAM. The pDR is also an integrating nephelometer (scattering coefficient range = 1.5 × 10^{-6} to 0.6 m^{-1} at wavelength 880 nm). It records in units of μg/m^3, as calibrated by the manufacturer using a fine ISO test dust (specific gravity, 2.6; index of refraction = 1.5−0*i*, $d_{p50} = 2-3$ μm). Readings can be converted to light scattering units using the following equation: μg/m^3 reading = 1.023 × (light scatter in 1×10^{-3} m^{-1}) (Thermo-MIE technical note 8-A). As with nephs, the pDRs were programmed to record 10-min averages. The pDRs were operated at the passive mode without any size selective inlet. They were zeroed for 2 min before each monitoring session. This zeroing procedure was repeated daily during the monitoring period, using the

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zeroing bag with particle-free air through a HEPA filter supplied by the manufacturer.

Gravimetric Measurements. Personal PM mass concentrations were made with the small lightweight $HPEM_{2.5}$ (*18*, *19*) (Harvard University, Boston, MA). $HPEM_{2.5}$ is a single-stage inertial impactor with a 50% cut point of 2.4 ± 0.1 μm (*19*). Particles greater than 2.5 μm were collected on the impaction plate in the $HPEM_{2.5}$ immediately downstream to the inlet. The impaction plate was filled with silicon vacuum grease to reduce particle bounce (*19*). The indoor and outdoor 24-h average $PM_{2.5}$ mass concentrations were measured with a single-stage Harvard Impactor (Air Diagnostics and Engineering, Inc., Naples, ME) (*20*). Airflow rates for the $HI_{2.5}$ (10 L/min) and $HPEM_{2.5}$ (4 L/min) were calibrated with a digital piston flowmeter (Drycal, DC-Lite, BIOS, Butler, NJ). All gravimetric $PM_{2.5}$ samples were collected on Teflon filters (Teflo with ring, 37 mm, 2 μm pore size, PTFE membrane; Gelman Laboratories, Ann Arbor, MI) over 24-h periods. The filters were conditioned at 22 ± 1.8 °C and 33 ± 2.5% RH for at least 24 h in an environmental chamber (*21*) before weighing with a Mettler Toledo ultramicrobalance (model UMT2).

The limit of detection (LOD), calculated as 3 times the standard deviation of field blanks deployed at 10% of total sample size, is 1 $\mu g/m^3$ for $HI_{2.5}$ and 4 $\mu g/m^3$ for $HPEM_{2.5}$, respectively. The 4 $\mu g/m^3$ LOD for $HPEM_{2.5}$ is similar to that reported by Sarnat et al. (*22*). The $HI_{2.5}$ measurements correlated well with the federal reference method for $PM_{2.5}$ ($FRM_{2.5}$) with a mean difference of 0.4 $\mu g/m^3$ and a coefficient of determination of 0.99 ($N = 55$ collocated pairs) at an outdoor fixed site. The mean difference between the $HPEM_{2.5}$ and $FRM_{2.5}$ measurements was 0.7 $\mu g/m^3$ ($R^2 = 0.93$, $N = 25$ pairs). The mean difference between the $HI_{2.5}$ and $HPEM_{2.5}$ was 0.1 $\mu g/m^3$ ($R^2 = 0.96$, $N = 17$ pairs). Duplicate $HI_{2.5}$ and $HPEM_{2.5}$ were also deployed at 10% total sample size for estimating precision.

Quality Control. All data were examined for irregularity and noncompliance with our standard operating procedures. "Bad" samples were flagged and removed from the analyses. The "bad" neph samples (4.6% of total samples) included values below -0.0001 m^{-1} (unreasonable values) and above 0.006 m^{-1} (~2145 $\mu g/m^3$; any values $\geq$ 0.006 m^{-1} are off-scale and likely due to neph malfunctions), taken during the period of fan or heater malfunctions, outdoor measurements taken without rain covers, or measurements with unexpected averaging time intervals during the field collocation tests. For pDR measurements, the "bad" samples (<1% of total samples), included extreme outliers greater than 1 mg/m^3 that were at least 25 times greater than the previous and subsequent 10-min readings. Such outliers were due to instrument artifacts after we examined the data and the subjects' home activity logs. Gravimetric samples were removed from the data when the flow rates fell beyond 10% of the designated flow rates. Bad samples removed from the $HI_{2.5}$ and $HPEM_{2.5}$ measurements (~6.4%) were largely due to pump or battery failure. The final sample size for each measurement analyzed in this paper is shown in Table 1.

Data Analysis. Precision of the gravimetric monitors, estimated from the paired collocated monitors, was defined as the root-mean-square error (i.e., square root of the average of the variance of the difference between the pairs divided by the mean). Uncertainty and precision of the light scattering devices were determined by the collocation tests of all devices. Uncertainty was defined as the square root of the average of the variance of 10-min intervals divided by the number of collocated instruments. Uncertainty was divided by the mean measurement (and presented in percentage) to obtain the precision. For subsequent data analysis, monitoring days were grouped into either "cooking" or "noncooking" days. A monitoring day was designated as a "cooking" day when at least one meal was prepared in the home during a 24-h monitoring period. Generalized Estimating Equations (GEE) in SAS program (release 8.02, SAS Institute Inc., Cary, NC), which control for autocorrelation within site, were used to compare measurements between different types of instruments, stratified by cooking days. Effects of potential sources of PM from daily household activities on the performance of the pDR and neph were further examined. These household activities, used as the independent variables, included cooking in general; using the stove for baking, frying, grilling, and sautéing; other cooking activities, including using the microwave or boiling water; opening windows; cleaning (vacuuming, dusting, sweeping, wiping, etc.); using air-cleaning devices; and having pets in the house. We converted neph readings to $PM_{2.5}$ concentrations using the regression equations as shown in Figure 4a (presented later) to calculate the differences between neph and pDR measurements.

Results

Precision. The precision for the collocated HI monitors was 7%, with a mean difference of 0.2 $\mu g/m^3$ (paired *t*-test, $p = 0.08$, $N = 107$) between the collocated $HI_{2.5}$ samples. The precision for the $HPEM_{2.5}$ was 22%, with a mean difference of 0.2 $\mu g/m^3$ (paired *t*-test, $p = 0.64$, $N = 46$) between the collocated $HPEM_{2.5}$ carried by technicians. This higher imprecision for the $HPEM_{2.5}$ versus the HI was due to higher uncertainty associated with the smaller flow rate and subsequent smaller mass loading.

The precision of the nephs, based on the lab collocation Test 6a, ranged between 3% and 8% at mean B_{sp} values between 0.392×10^{-5} and 1.024×10^{-5} m^{-1} (or 1.4 and 3.6 $\mu g/m^3$, levels not uncommonly found at the home sites in our study). The uncertainty ranged between 0.022×10^{-5} and 0.039×10^{-5} m^{-1} (or 0 and 0.1 $\mu g/m^3$). All nephs were highly correlated with a Pearson's correlation coefficient greater than or equal to 0.97. Figure 1 shows the agreement among the nine nephs (10-min averages) during one 10-day session.

The uncertainty of the PDRs ranged between 0.8 and 1.5 $\mu g/m^3$ during the lab collocation tests at PM levels between 1.5 and 3.8 $\mu g/m^3$. At these low PM levels, the pDR precision ranged between 40% and 53%. In our collocation tests, the pDRs were not necessarily zeroed in a same "zeroing bag" to simulate actual field operations and to account for bias due to such potential zeroing operation error. Thus, the reported uncertainty and precision represent actual operation and instrument errors. When the zeroing bias was removed, the uncertainty (ranging between 1.3 and 0.5 $\mu g/m^3$) and precision (ranging between 18% and 35%) were reduced and represented unbiased instrument errors. Table 2 shows the bias unadjusted and adjusted uncertainty and precision of the pDR from the field tests inside a residence, grouped by PM concentrations. Most pDR measurements at the residence were below 40 $\mu g/m^3$, with an adjusted uncertainty ~1 $\mu g/m^3$. As PM concentration increased, the uncertainty increased, while the imprecision decreased. The high uncertainty at PM > 200 $\mu g/m^3$ was due to the proximity of one or two of the pDRs to local sources (e.g., burning incense). The average precision was 12% for unadjusted pDR measurements and was reduced to 6% after removing the zeroing bias. The overall adjusted uncertainty was 4.4 $\mu g/m^3$. The Pearson's correlation coefficient between these collocated pDRs ranged between 0.95 and 1.00.

PDR versus Nephelometer. The pDRs were collocated with nephs in both laboratory and residential settings. In the laboratory, pDRs did not correlate well among themselves nor with nephs, due to the extremely low PM levels. At higher levels in the field test (Test 5bii), the Pearson's correlation coefficient (*r*) for the three pDRs and the neph was greater than or equal to 0.95 (Figures 2a). The coefficient of

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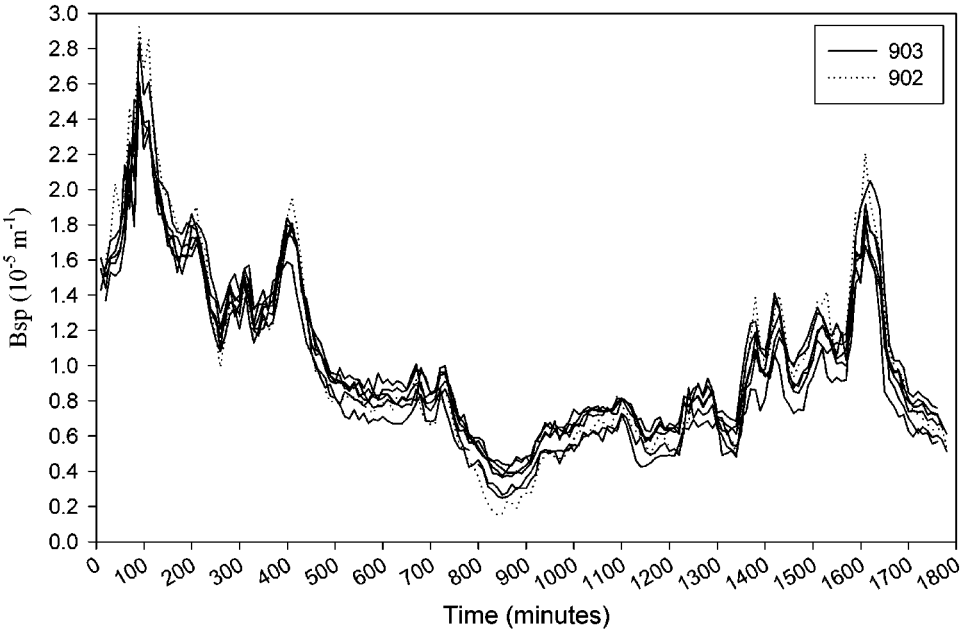


FIGURE 1. Laboratory collocated Radiance nephelometers (shown in 10-min resolution from session S7).

TABLE 2. Summary of the Field Collocated Test for Three Thermo-MIE PDRs (10-min measurements) at One Residence

range ($\mu g/m^3$)	*N*	mean ($\mu g/m^3$)	SD ($\mu g/m^3$)	uncertainty ($\mu g/m^3$)	precision	adjusted uncertainty ($\mu g/m^3$)[a]	adjusted precision
0 ≤ pDR ≤ 10	288	6.8	2.2	1.6	24%	0.8	12%
10 < pDR ≤ 20	604	15.6	2.4	1.8	11%	0.8	5%
20 < pDR ≤ 30	352	24.2	2.6	2.0	8%	1.2	5%
30 < pDR ≤ 40	101	34.4	3.2	2.5	7%	1.3	4%
40 < pDR ≤ 50	56	44.1	4.2	3.6	8%	2.9	7%
50 < pDR ≤ 60	22	53.6	4.1	3.1	6%	2.1	4%
60 < pDR ≤ 70	15	64.9	4.2	3.1	5%	1.9	3%
70 < pDR ≤ 100	35	84.7	7.2	7.3	9%	6.9	8%
100 < pDR ≤ 200	53	137.7	9.0	7.2	5%	6.2	4%
pDR > 200	40	382.4	22.7	20.3	5%	19.6	5%
overall	1566	34.2	3.4	5.3	12%	4.4	6%

[a] Adjusted for zeroing bias.

determination between pDRs and nephs did not vary with the pDR zeroing frequency (i.e., never, once per day, and once per session). However, the pDR3 that was only zeroed once per session in a separate zero bag was consistently 4 $\mu g/m^3$ higher than the other two pDRs throughout the 11-day test period (Figure 2b). A later validation test at the same home showed comparable yet lower coefficients of determination ($R^2 = 0.90-0.93$).

When pDRs were collocated with nephs inside the subjects' homes, the mean coefficient of determination decreased to 0.87 (10-min averages, Table 3). The coefficient of determination varied significantly by home, ranging between 0.75 and 0.96 for 10-min averages. Examining the data further revealed that particles emitted from cooking activities resulted in different responses in the instruments. Figure 3 shows the ratio of the pDR to neph reading at one residence. The ratio was generally stable ($\sim$0.60 g/m^2) but spiked during cooking hours (around 9 a.m.) on most days. This indicated that during cooking episodes, pDRs exhibited much greater response than nephs. We examined how water vapor might be responsible for this result but did not find a significant RH change during this time.

Light Scattering Devices versus Gravimetric Monitors. Both the neph and pDR correlated well with the 24-h integrated gravimetric measurements, with higher coefficients of determination for the neph than for the pDR. The higher coefficient of determination between indoor neph and $HI_{2.5}$ on cooking days (0.93) as compared to noncooking days (0.81) is probably due to the higher variability in PM levels on cooking days (Figure 4a). Note that without the highly influential point at $HI_{2.5} \sim 50$ $\mu g/m^3$, the coefficient of determination was only slightly reduced to 0.89 (slope = 0.27). Nephs did not exhibit significantly different responses to $PM_{2.5}$ mass concentrations on cooking and noncooking days. Furthermore, the relationship between neph responses

TABLE 3. Regression Results for Collocated pDR and neph Measurements at Subjects' Homes at 10-min Intervals (neph Measurements Were Used as the Independent Variable)

site[a]	*N*	intercept ($\mu g/m^3$)	slope ($\mu g/m^2$)	R^2	neph model
F18	707	0.12 (0.13)[b]	4.65 (0.08)	0.83	902
F43	988	4.09 (0.11)	2.46 (0.04)	0.77	902
F47	1133	1.50 (0.08)	2.80 (0.06)	0.68	902
S46	1132	1.72 (0.44)	7.09 (0.06)	0.93	903
S58	1326	3.08 (0.14)	4.04 (0.05)	0.84	903
S62	279	0.96 (0.45)	5.09 (0.06)	0.96	903
S70	1134	−1.54 (0.45)	6.62 (0.11)	0.75	903
overall	6705	1.47 (0.12)	4.64 (0.52)	0.87	

[a] Sites F35 and S65 had model 902 nephelometers whose values could not be corrected because no lab or field collocation data were available. These two sites were excluded when calculating the overall values. [b] Number in parentheses is standard error.

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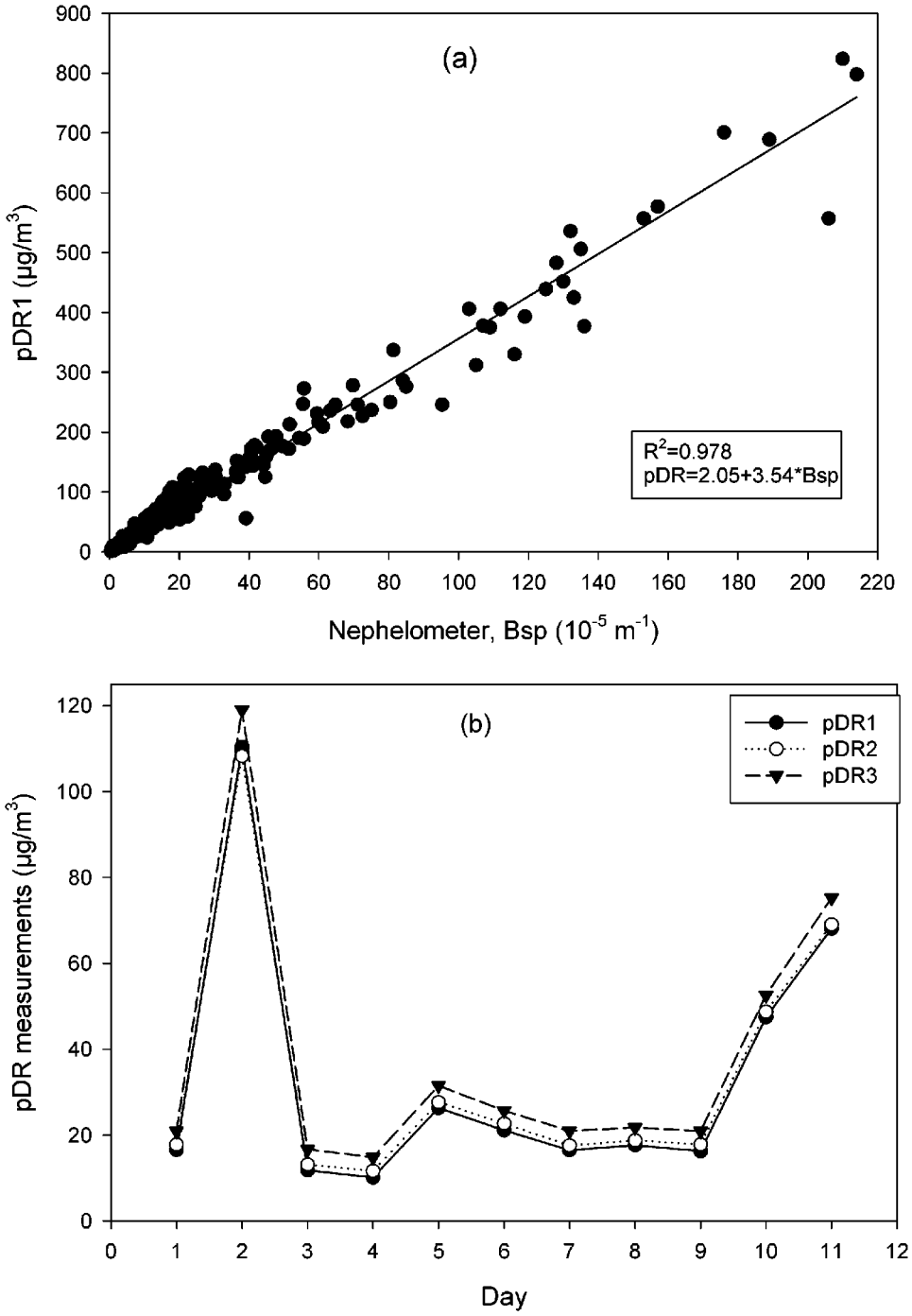


FIGURE 2. Correlation (a) between collocated pDR and nephelometer measurements and (b) among the 3 pDRs at one home Dec 17−29, 1999 (10-min averages, *N* = 1566).

and $HI_{2.5}$ was similar across homes (i.e., comparable regression slopes) on both cooking and noncooking days. Neither the number of cooking hours nor the number of meals cooked per day significantly affected the regression slopes for neph versus $HI_{2.5}$ in cooking homes. At home outdoor sites, the coefficient of determination for the neph versus $HI_{2.5}$ was 0.88 (Figure 4b). As expected, there was no distinction in responses between cooking and noncooking days (p = 0.48).

Indoor pDRs correlated well (but to a lesser extent as compared to nephs) with the $HI_{2.5}$ at residences (Figure 5a). Similar regression slopes were found between pDRs and $HI_{2.5}$ on both cooking and noncooking days. Moreover, the pDR response curve did not differ significantly among homes (p = 0.11, GEE). The pDR measurements generally overestimated gravimetric $PM_{2.5}$ measurements, with a mean indoor pDR/$HI_{2.5}$ ratio of 1.56 ± 0.07. This ratio is relatively stable for the entire dynamic range of concentrations, except for very low concentrations. As the pDR uses a calibration dust density of 2.6 g/cm^3, overestimating gravimetric measurements by a factor of 1.56 gives an estimated indoor particle density of 1.67 g/cm^3. This estimated indoor particle density is within the range for sulfates (1.76), nitrate (2.1), organic carbon (1.4), and elemental carbon (2.0) (*23*). Somewhat different results were obtained for the comparison with the personal gravimetric measurements. When both the pDR and $HPEM_{2.5}$ were carried by subjects, a smaller coefficient of determination (0.48) was observed (Figure 5b). Again, the regression slope did not differ significantly ($p > 0.05$) between cooking and noncooking days. The mean personal pDR/

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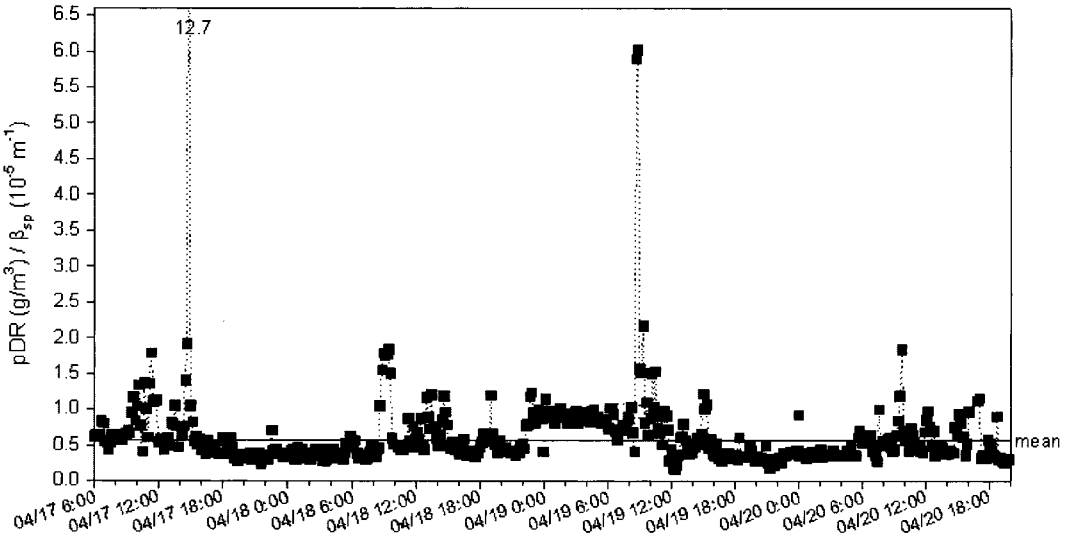


FIGURE 3. Cooking spikes of the ratio of pDR to nephelometer measurements (g/m²) at one residence, S58 (group home).

$HPEM_{2.5}$ ratio is 1.27 ± 0.08. This overestimation factor leads to an estimated density of the personal aerosol of 2.05 g/cm^3. This larger density seem reasonable if the personal aerosol is largely suspended crustal material, as has been found by Ozkaynak et al. (24). These overestimation factors were similar but slightly lower than the mean indoor $pDR/HI_{2.5}$ ratio 1.41 reported by Quintana et al. (*7*) and the mean personal pDR/PEM ratio 1.49 by Howard-Reed et al. (*6*).

Concurrent neph, pDR, $HI_{2.5}$, and HPEM measurements were taken at two sites. The comparisons among monitors are shown in Figure 6. The regression results showed that the neph light scattering response (slope ~ 0.29) is identical when compared with the HI (shown as filled triangles in figure) or HPEM (shown as filled squares) measurements. These slopes are identical to those obtained in Figure 4, parts a and b. Neph agreed better with the pDR (shown as filled circles, $R^2 = 0.94$) than with HPEM ($R^2 = 0.85$). When the pDR measurements were corrected for particle density with a factor of 1.56 (shown as open circles), they lie close to the gravimetric measurements. The pDRs have higher uncertainties than nephs and also the HPEM is less precise than $HI_{2.5}$. Therefore, the coefficient of determination observed for pDR versus $HI_{2.5}$ (0.89) was greater than that for pDR versus $HPEM_{2.5}$ (0.78) (not shown in Figure 6).

Factors Affecting Responses of the pDR and Neph. Relative humidity has been long reported to play an important role in particle volume and its associated light scattering, thus affecting the performance of the light scattering-based devices (*7*, *17*, *25*). Quintana et al. (*7*) observed a dramatic increase in particle readings for the pDRs at RH greater than 85%. However, RH in indoor environments normally does not exceed 70% and thus should not be a major factor. In our study, the RH in the 40 monitoring homes varied between 25% and 64% (mean $\pm$ SD $= 41 \pm 8\%$). Only 0.9% of the monitoring days observed RH values greater than 60%. Although cooking homes exhibited a slightly higher average RH value ($44 \pm 8\%$), it was not significantly higher than that in the noncooking homes ($38 \pm 8\%$). Regression analysis for indoor RH versus pDR measurements indicated that the pDR was not associated with RH inside the study residences. However, indoor neph measurements were weakly but significantly affected by RH ($R^2 = 0.02$, $p < 0.05$). Regression results show that a 25% difference in RH corresponds to a difference of 0.75×10^{-5}m^{-1} in B_{sp} or approximately 2.6 μg/m^3 in $PM_{2.5}$. Adjusting for RH did not alter the association between indoor nephs and $HI_{2.5}$, nor did it change the responses in the cooking or noncooking models shown in Figure 4a.

Effects of PM sources from household activities on the responses of the pDR and neph were further examined at the nine homes where concurrent pDR and neph measurements were taken. Results from the GEE model using the 10-min pDR and converted neph measurements (both in μg/m^3) are summarized in Table 4. These potential PM-generating activities, including cooking and cleaning, comprised a very small percentage (2.3%) of the overall sampling time. When there were no such activities, the difference between the pDR and neph measurements was on average 5.8 μg/m^3. "General cooking or stove use" were significantly ($p < 0.0001$) associated with an increase in the difference between the pDR and neph measurements. Baking and frying were the two most commonly reported uses of the stove, and both activities caused an immediate increase in the difference between pDR and neph readings. When baking occurred, the pDR read on average 36.1 μg/m^3 higher than the neph, the largest differences observed during cooking. Cleaning did not cause a significant change in the responses between pDR and neph.

Discussion

Our results indicate that the neph is more precise than the pDR, especially at low PM concentrations. During the field tests at one residence where there were no cooking or strenuous household activities (Test 5bii), the pDR and neph agreed very well ($R^2 > 0.90$). However, when deployed at various homes with regular household activities (Test 5bi), the real-time correlation between the pDR and neph depends on the nature of PM emissions from these activities ($R^2 = 0.75-0.96$). The more heterogeneous the PM source types within the household, the less the correlation between the neph and pDR. The most important household activity that significantly elevates pDR light scattering responses, while not affecting neph light scattering responses, was cooking, in particular, baking or frying.

However, this cooking effect was not observed when the 24-h integrated gravimetric $PM_{2.5}$ measurements were compared with the 24-h average neph or pDR measurements. We were obligated to use the rough criterion of whether one or more of the meals were cooked on 1 day to divide data into cooking and noncooking days. To more precisely analyze

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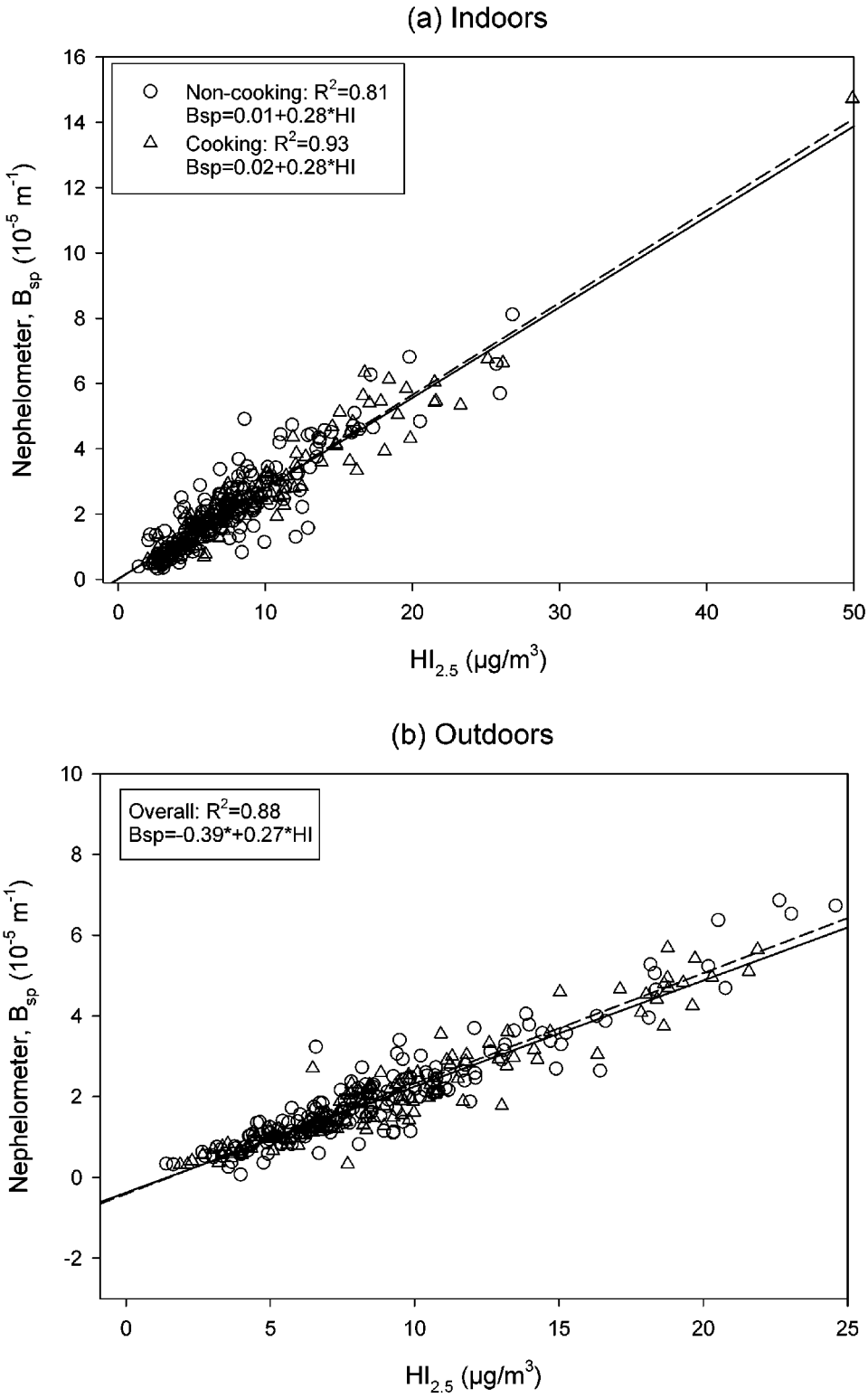


FIGURE 4. Collocated nephelometer and $HI_{2.5}$ at cooking and noncooking homes: (a) indoors and (b) outdoors.

the effects of cooking on the light scattering response, it would be ideal to divide the data into cooking hours and noncooking hours. However, it would not be possible to compare these real-time light scattering devices against the conventional impactor measurements during cooking, as the impactor measurements were integrated over time. It is possible that most of the "cooking days" have a preponderance of hours when there was no associated cooking aerosol. This may result in the observed stability of the neph versus HI relationship on a 24-h basis, independent of cooking activities.

The regression slope for $HI_{2.5}$ versus neph indoors is not different from that outdoors. Our outdoor $HI_{2.5}$ versus neph response slope (2.8 m^2/g) agrees with the results of Waggoner and Weiss (*15*), who observed similar slopes at five outdoor, western U.S. sites with different surrounding sources (i.e., urban, industrial, rural, or residential) and relative humidities. This low variability was attributed to similarities in the average

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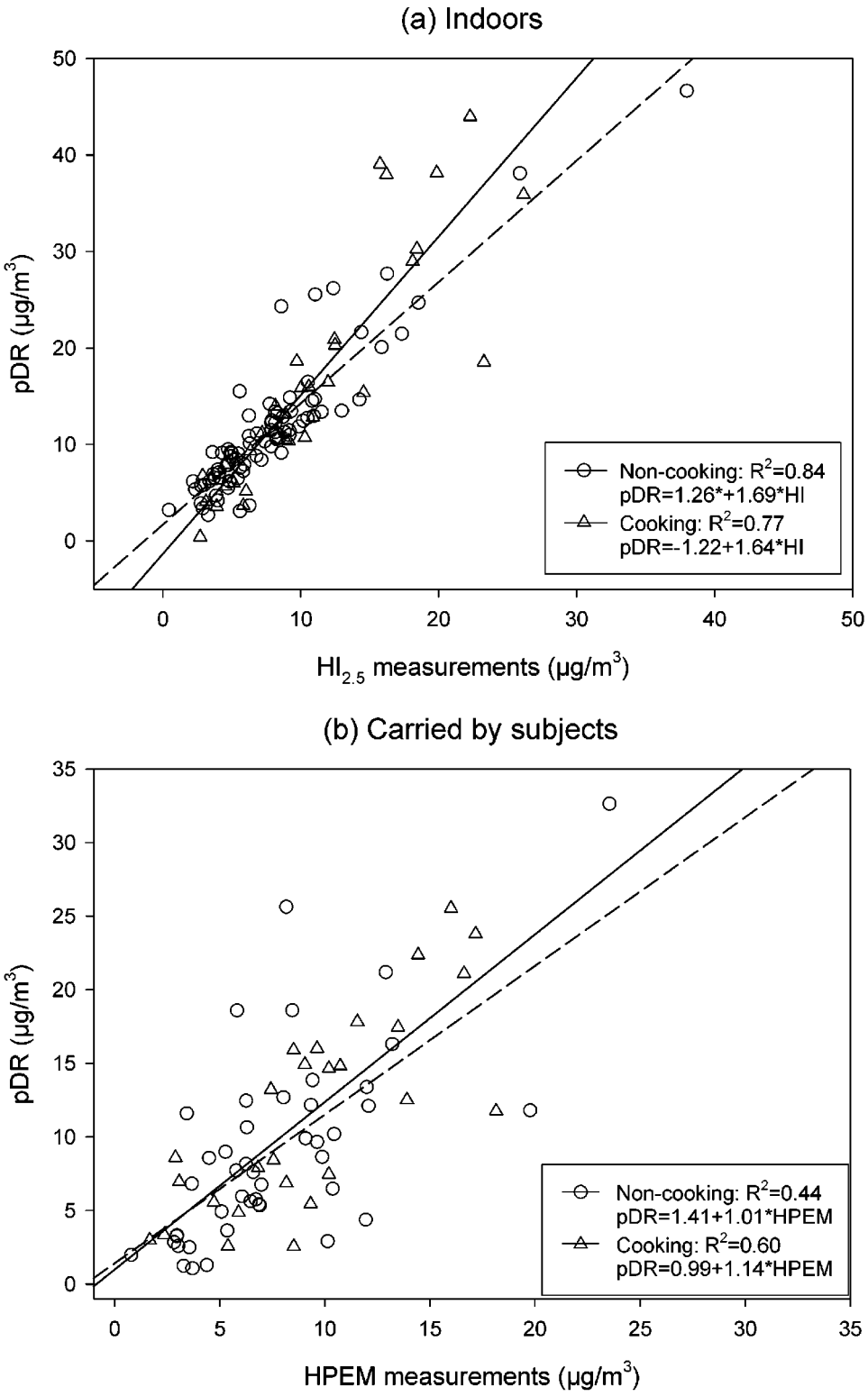


FIGURE 5. (a) Collocated pDR and $HI_{2.5}$ at 16 residences and (b) collocated pDR and $HPEM_{2.5}$ carried by subjects, grouped by cooking and noncooking activities.

aerosol size distribution at each location and the fact that their nephelometer inlet was heated. The slope is fairly constant as long as RH is below 60% (*25*).

Abt et al. (*26*) have shown the importance of indoor sources, including cooking, cleaning, and movement of people, and air exchange rates on observed indoor concentrations. When there is a separate indoor source of particles from cooking, there is additional mass above 1 μm. Different types of cooking, such as baking, frying, toasting, and so forth, resulted in different particle size distribution (*27*). Wallace (*28*) has shown that gas burners, baking, and toasting generate particles in the size range of 0.01−0.02 μm at the peak number concentration, while frying and broiling produce peaks ranging from 0.05−0.1 μm. Other research indicates that cooking activities including baking, sauteing, toasting, and frying primarily generate particles in the range of 0.02−0.1 μm (*29*). As the neph and pDR have different light scattering wavelengths (530 vs 880 nm) and optimized

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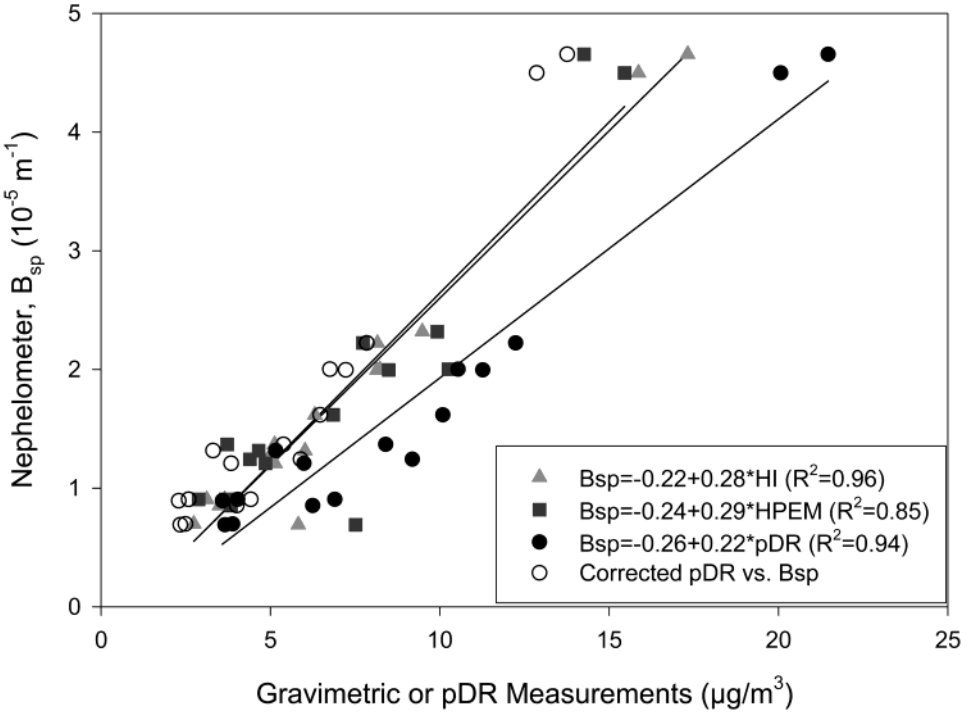


FIGURE 6. Collocated nephelometer, pDR, $HI_{2.5}$, and $HPEM_{2.5}$ inside residences F47 and S58.

TABLE 4. Variation in the pDR−neph Difference (μg/m^3) Based on Whether or Not a PM-Generating Activity Was Currently Occurring (10-min Resolution)

	activity did not occur			activity occurred			
PM-generating activity	min	difference	std error	min	difference	std error	*p*-value[a]
general cooking or stove use[b]	65 610	5.74	0.17	1450	12.8	2.12	<0.0001
any stove use[c]	66 380	5.73	0.17	680	21.9	3.99	<0.0001
stove used for baking	66 910	5.82	0.18	150	36.1	10.5	<0.0001
stove used for frying	66 780	5.80	0.17	280	28.6	6.83	<0.0001
stove used for sautéing	66 970	5.89	0.18	90	4.93	3.42	0.84
other or unknown stove use	66 900	5.89	0.18	160	6.20	3.20	0.93
cleaning	66 980	5.90	0.18	80	2.32	0.68	0.48

[a] Test results for the two differences in the pDR−neph when activities occurred or did not occur. [b] "General cooking or stove use" includes any cooking, "any stove use" shown below except microwave, as recorded on the time−activity diaries. [c] "Any stove use" includes frying, grilling, sautéing, broiling, or baking, as recorded on daily follow-up questionnaire.

detecting particle sizes (0.1 vs 0.1−10 μm), the overall scattering efficiency of the two devices may well vary as the particles infiltrating from outdoors are mixed with the cooking-generated particles. When most cooking aerosols are in the coarse mode, the pDR's response would increase while the neph's would not, resulting in the observed increase in the differential response between the pDR and neph measurements during these cooking periods (Table 4). This overall result is reversed if particles generated during cooking are between 0.1 and 0.3 μm. In this latter case, the pDR scattering efficiency would decrease with cooking activity because these smaller particles are not detected by the pDR. Alternatively, if the cooking activities generate particles between 0.3 and 0.7 μm, then both instruments will scatter light efficiently and their mass scattering efficiencies would not change relative to noncooking periods.

Our neph versus $HI_{2.5}$ results are similar to those reported by Brauer (*5*), who collocated unheated nephs (without any $PM_{2.5}$ inlet sizers) with $HI_{2.5}$ and HI_{10} samplers and reported a high correlation between indoor neph and $HI_{2.5}$ measurements ($R^2 = 0.79-0.96$). Quintana et al. (*7*) reported a much lower coefficient of determination of 0.42 for collocated indoor pDR (without a $PM_{2.5}$ sizer) and $HI_{2.5}$. Neither study identified effects of indoor sources on the performance of the light scattering devices. Our observation of a decrease in the coefficient of determination for pDR and $HPEM_{2.5}$ (0.48) when both monitors were carried by subjects is most likely due to the increased heterogeneity of PM sources encountered and the inherently less precise gravimetric measurements of the HPEM. It is not clear to what degree the 22% imprecision in personal $HPEM_{2.5}$ measurements may affect these between-method comparisons. This finding, however, is within the range reported by Quintana et al. (*7*) ($R^2 = 0.42$ for indoor $HI_{2.5}$ vs pDR without $PM_{2.5}$ inlet size cut) and Howard-Reed et al. (*6*) ($R^2 = 0.66$ for personal PEM vs pDR without $PM_{2.5}$ inlet size cut). Zero drift was not observed among these pDRs during the 11-day test period, indicating that the frequency of pDR zeroing probably did not affect the performance of the pDRs. It is more important to simultaneously zero these pDRs in a same zero bag to avoid baseline bias.

In conclusion, both nephs and pDRs correlated well with the integrated gravimetric instruments when placed at fixed locations in residential environments, with nephs exhibiting a higher coefficient of determination than pDRs. On a 24-h basis, the mass scattering efficiency of the nephs and the pDRs did not differ between residences, nor between homes with or without cooking activities, nor between indoor vs outdoor locations. On a 10-min basis, both light scattering devices generally compared well with each other at fixed

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indoor locations, except during the relatively brief periods of indoor source activity, primarily cooking. The correlations with gravimetric mass for the pDRs worn by subjects was significantly lower than at the fixed locations. Both devices are potentially useful in future exposure assessment and health effects studies when 24-h time-weighted values are used or when peak values are interpreted carefully with the assistance of other PM related information such as indoor and personal activities.

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